BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 254 · Von Neumann and Harvard architectures

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the ideal completion cycle for four independent instructions in a three-stage pipeline?

Q2. What does MDR contain during an instruction fetch?

Q3. What happens in the Decode stage?

Q4. Calculate Sum and Carry-out for a full adder with A=1, B=0, Carry-in=1. Explain in binary.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Von Neumann — Architecture with shared memory space for instructions and data.

Harvard — Architecture with separate program and data storage paths.

bottleneck — Limit caused by a shared resource or transfer path.

instruction cache — Fast cache storing instructions.

hybrid architecture — Combines features of different architectural approaches.

New learning

Von Neumann organisation — Instructions and data share the same address space and a pathway; access can compete for bandwidth.

Harvard organisation — Instruction and data memories/address spaces are separate, allowing independent access paths.

Recognise contemporary designs — Many processors have aspects of both, such as a unified main memory and separate instruction/data caches (modified Harvard).

Worked example

A CPU needs to fetch an instruction while reading its operand. Explain possible contention in a Von Neumann design.

1. Predict first — The instruction fetch and data read may both need the same memory transfer pathway.

2. Explain and check — The requests may have to be scheduled rather than served simultaneously; this is related to the Von Neumann bottleneck.

We Do · Practise with guidance

A microcontroller uses separate program and data memories. Identify the architecture and a useful consequence.

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Program ROM —instruction path→ CPU
Data RAM    ⇄data path⇄ CPU
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Predict first

Separate program and data memories indicate Harvard-style organisation.

Explain and check

Independent transfer pathways can allow concurrent instruction fetch and data access, subject to the processor design.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State where instructions and data are stored in Von Neumann and Harvard architectures.

Question 2

A CPU frequently fetches an instruction while reading data. Explain a potential Von Neumann limitation and Harvard advantage.

Question 3

A modern CPU uses unified main memory but separate L1 instruction and data caches. Explain why it cannot be described as a purely classical Harvard or purely classical Von Neumann design.

Exit Ticket · Answer before checking

Every modern processor is exclusively a pure Von Neumann or pure Harvard machine.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Cycle 6 (=4+3−1).

Q2. The instruction being transferred from memory.

Q3. The CU identifies and interprets the opcode/operands.

Q4. 1+0+1=2 decimal = 10_2, so Sum=0 and Carry-out=1.

You Do pages · Independent questions

Q1. [4] Von Neumann uses a shared instruction/data memory space [2]; Harvard has separate instruction/data memories or address spaces [2].

Q2. [6] Shared transfer pathway [1] used for both instructions and data [1], which can cause contention [1]; Harvard uses independent paths [1], potentially overlapping transfers [1] although overall speed depends on implementation [1].

Q3. [6] Unified main memory is shared [1], a Von Neumann feature [1]; separate L1 caches [1] offer separate instruction/data access paths [1], a Harvard-like feature [1]; this is a hybrid or modified Harvard organisation [1].

Exit Ticket · Model answer

False. Hybrid designs are common.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding